THE MAXIMUM NUMBER OF SQUARES OF AN $n \times n$ BOARD WITH NO TWO EXACTLY TWO STEPS APART IN A ROW OR COLUMN: YANG HONG'S CONJECTURE FOR OEIS A348484

MATH0

Abstract. Let $a(n)$ be the largest number of squares of an $n \times n$ chessboard that can be chosen so that no two chosen squares lie in the same row at column distance exactly 2, or in the same column at row distance exactly 2 (OEIS A348484). We prove the formulas conjectured in that entry: $a(n) = n^2/2$ for $n \equiv 0 \pmod 4$, $a(n) = (n^2+1)/2$ for odd n, and $a(n) = n^2/2 + 2$ for $n \equiv 2 \pmod 4$; equivalently $2a(n) = n^2 + \varepsilon(n)$ with $\varepsilon(n) = 0, 1, 4, 1$ for $n \equiv 0, 1, 2, 3 \pmod 4$. The matching lower bounds come from a 2×2-block checkerboard already described in the entry by R. J. Mathar; the new part is the upper bound, obtained from an explicit involution of the board that pairs squares two steps apart and has exactly $\varepsilon(n)$ fixed points. To our knowledge this is a new result, although an elementary one: it also follows from the independence number of grid graphs, and the OEIS entry still lists the formulas as conjectures. The theorem is verified in Lean 4 against a statement that math0 formalized from the OEIS entry, whose definitions are checked by the kernel against all ten listed terms; this statement is not part of the formal-conjectures repository.

1. Introduction

The OEIS sequence A348484 [5], contributed by Yang Hong on October 20, 2021, gives the maximum number $a(n)$ of squares of an $n \times n$ chessboard that can be chosen so that no two chosen squares are *two steps apart*, that is, in the same row with columns differing by exactly 2, or in the same column with rows differing by exactly 2 (Definition 1). Its listed terms are $1, 4, 5, 8, 13, 20, 25, 32, 41, 52$ for $n = 1, \ldots, 10$. The entry's FORMULA section records three conjectured formulas, one for each residue class of n modulo 4 with odd n combined (Section 2). We prove them for every $n \geq 1$ (Theorem 2).

Prior work. On December 3, 2022, R. J. Mathar added a comment to the entry: the sequence with generating function

$$\frac{-x\,(1 + 2x - 2x^2 + 2x^3 + x^4)}{(1+x)(1+x^2)(x-1)^3}$$

"is a lower bound for a(n) achieved by packing 2x2 squares with 1's and 2x2 squares with 0's in a checkerboard pattern into the chessboard." Its coefficients are exactly the conjectured values (Mathar's configuration, with the corner block filled with 1's, is the set S_0 of Definition 7, whose size is computed in Proposition 9 and Lemma 10). So the conjectured formulas were known to be lower bounds; the comment does not address an upper bound. The only cross-reference of the entry is A048716 [4], the numbers k with k AND $4k = 0$: a single row, read as the binary number $k = \sum_{j \in \text{row}} 2^j$, contains no two squares two steps apart exactly when k belongs to A048716. In revision #33 (December 6, 2022), which the referees found to be the current one, the formulas are still labelled "Conjectures", the keyword `more` is set, and no proof is recorded. The math0 referees found no proof of the upper bound on arXiv, MathOverflow, Mathematics Stack Exchange, the SeqFan mailing list, or in a general web search, and no other OEIS entry

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that cites A348484. The statement is not among the OEIS statements of the formal-conjectures repository [1].

We therefore classify the result as *new*, with a caveat: it is elementary. The "two steps apart" graph on the board is a disjoint union of four grid graphs, and the result follows from the standard fact that the grid graph $P_r \,\square\, P_s$ has independence number $\lceil rs/2 \rceil$; see Remark 11, where we also explain how our proof is this argument written without coordinates on the four grids.

2. Statement

Definition 1. Let $n \geq 0$, and let $B_n = \{0, \ldots, n-1\}^2$ be the set of squares of the $n \times n$ board, (i, j) lying in row i and column j. Squares $c = (i, j)$ and $d = (i', j')$ are *two steps apart* if

$$\big(i = i' \text{ and } |j - j'| = 2\big) \quad \text{or} \quad \big(j = j' \text{ and } |i - i'| = 2\big).$$

A set $S \subseteq B_n$ is *admissible* if no two of its squares are two steps apart, and $a(n)$ is the largest size of an admissible set (the empty set is admissible, so $a(n)$ is well defined; $a(0) = 0$).

The entry A348484 [5], authored by Yang Hong (October 20, 2021), has offset 1 and name "Maximum number of squares on an n X n chessboard such that no two are two steps apart horizontally or vertically." Its FORMULA section (revision #33, December 6, 2022) contains the following unsigned lines, verbatim:

```
Conjectures:
a(n) = n^2/2 for n == 0 (mod 4).
a(n) = (n^2 + 1)/2 for n == 1 or 3 (mod 4).
a(n) = n^2/2 + 2 for n == 2 (mod 4).
```

Theorem 2. *For every $n \geq 1$: $a(n) = n^2/2$ if $n \equiv 0 \pmod 4$, $a(n) = (n^2+1)/2$ if $n \equiv 1$ or $3 \pmod 4$, and $a(n) = n^2/2 + 2$ if $n \equiv 2 \pmod 4$. Equivalently, for every $n \geq 1$,*

$$2a(n) = n^2 + \varepsilon(n), \qquad \varepsilon(n) = \begin{cases} 0, & n \equiv 0 \pmod 4, \\ 1, & n \equiv 1 \text{ or } 3 \pmod 4, \\ 4, & n \equiv 2 \pmod 4. \end{cases}$$

The identity $2a(n) = n^2 + \varepsilon(n)$ also holds, trivially, for $n = 0$, and the proof in Section 3 does not use $n \geq 1$; the theorem, like the formal statement below, concerns $n \geq 1$ only, matching the offset of the entry.

Lean formalization. The statement was formalized by math0 in the file `lean/Math0/Statements/OEIS/A348484.lean`, and was reviewed for fidelity and frozen (SHA-256 `6dc8c7d5...1a045c66`) before any proof attempt. Its definitions and the statement are, verbatim except that the file header (copyright notice and `import Mathlib`), docstrings, blank lines and the data-check definitions and lemmas discussed below are omitted:

```
namespace Math0.Statements.A348484
def TwoApart {n : ℕ} (c d : Fin n × Fin n) : Prop :=
  (c.1 = d.1 ∧ (c.2.val + 2 = d.2.val ∨ d.2.val + 2 = c.2.val)) ∨
    (c.2 = d.2 ∧ (c.1.val + 2 = d.1.val ∨ d.1.val + 2 = c.1.val))
instance {n : ℕ} (c d : Fin n × Fin n) : Decidable (TwoApart c d) :=
  inferInstanceAs (Decidable (_ ∨ _))
def Admissible {n : ℕ} (S : Finset (Fin n × Fin n)) : Prop :=
  ∀ c ∈ S, ∀ d ∈ S, ¬ TwoApart c d
instance {n : ℕ} (S : Finset (Fin n × Fin n)) : Decidable (Admissible S) :=
  inferInstanceAs (Decidable (∀ c ∈ S, ∀ d ∈ S, ¬ TwoApart c d))
def a (n : ℕ) : ℕ :=
```

```
  ((Finset.univ : Finset (Finset (Fin n × Fin n))).filter Admissible).sup Finset.card
def statement : Prop :=
  ∀ n : ℕ, 1 ≤ n →
    (n % 4 = 0 → a n = n ^ 2 / 2) ∧
    (n % 4 = 1 ∨ n % 4 = 3 → a n = (n ^ 2 + 1) / 2) ∧
    (n % 4 = 2 → a n = n ^ 2 / 2 + 2)
end Math0.Statements.A348484
```

A square is an element of `Fin n × Fin n` (row, column), and `TwoApart` is Definition 1 with $|x - y| = 2$ written as $x + 2 = y \vee y + 2 = x$. `a n` is the supremum of the cardinalities of all admissible subsets of the board, i.e. their maximum, since $\emptyset$ is admissible. The statement quantifies over all $n \geq 1$, matching the offset. In it, `/` is division in $\mathbb{N}$, which is exact in each case (n^2 is even for even n, $n^2 + 1$ is even for odd n), so the three clauses mean the same as the entry's rational expressions; the three hypotheses cover every residue of n modulo 4.

The reading of "two steps apart horizontally or vertically" is confirmed by the entry's examples: for $n = 2$ all four squares are chosen, so diagonal neighbours are allowed; the $n = 4$ example `1100/1100/0011/0011` contains squares two steps apart diagonally; and the $n = 5$ example row `11001` contains squares three and four steps apart in a row.

Data check. The definition of `a` enumerates all 2^{n^2} subsets of the board, so the kernel can evaluate it directly only for very small n. The statement file proves

```
theorem a_eq_of_cert {n v : ℕ} (S : Finset (Fin n × Fin n)) (hS : Admissible S)
    (hcard : S.card = v) (lab : Fin n × Fin n → ℕ) (hlt : ∀ c, lab c < v)
    (hfib : ∀ c d, lab c = lab d → c = d ∨ TwoApart c d) : a n = v
theorem a_data : (List.range 10).map (fun i => a (i + 1)) =
    [1, 4, 5, 8, 13, 20, 25, 32, 41, 52]
```

The lemma `a_eq_of_cert` is a general certificate principle: an admissible set of v squares gives $a(n) \geq v$, and a labelling of the squares by labels $< v$ in which two distinct squares with the same label are two steps apart gives $a(n) \leq v$, because the labelling is injective on any admissible set. For each $1 \leq n \leq 10$ the file contains such a certificate, a witness given by row bitmasks and a label table, both found by computer from a maximum matching of the "two steps apart" graph, and its four side conditions are checked by `decide +kernel`, i.e. by evaluation in the Lean kernel. For $n = 3$, for example, the witness and the labels are (column 0 on the left)

$$\begin{array}{ccc} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{array} \qquad \begin{array}{ccc} 0 & 1 & 3 \\ 2 & 4 & 2 \\ 0 & 1 & 3 \end{array}$$

so the five labels $0, \ldots, 4$ show $a(3) \leq 5$, and the five chosen squares show $a(3) \geq 5$. The theorem `a_data` assembles the ten values, which agree with all ten listed terms of A348484. Independently, `a_small` evaluates $a(1), a(2), a(3)$ by brute force from the definition, also with `decide +kernel`. These lemmas validate the definitions only; the proof in Section 3 does not use them.

3. Proof

The proof follows the Lean file `lean/Math0/Frontier/OEIS/A348484.lean`; with each step we name the corresponding Lean declaration. Fix $n \geq 0$.

Definition 3 (`f`)**.** Let $f\colon \mathbb{N} \to \mathbb{N}$ be given by $f(j) = j + 2$ if $j \bmod 4 \in \{0, 1\}$ and $f(j) = j - 2$ if $j \bmod 4 \in \{2, 3\}$.

Lemma 4 (`f_f`, `f_cases`)**.** *For every $j \in \mathbb{N}$, $f(f(j)) = j$ and $|f(j) - j| = 2$; in particular $f(j) \neq j$.*

Proof. If $j \bmod 4 \in \{0,1\}$ then $f(j) = j+2$ has residue 2 or 3, so $f(f(j)) = j$. If $j \bmod 4 \in \{2,3\}$ then $j \geq 2$, so $f(j) = j-2$ is a genuine difference in $\mathbb{N}$, and it has residue 0 or 1, so $f(f(j)) = j$. In both cases $|f(j) - j| = 2$. □

Definition 5 (`sig`)**.** Define $\sigma\colon B_n \to B_n$ by

$$\sigma(i,j) = \begin{cases} (i, f(j)) & \text{if } f(j) < n, \\ (f(i), j) & \text{if } f(j) \geq n \text{ and } f(i) < n, \\ (i,j) & \text{if } f(j) \geq n \text{ and } f(i) \geq n. \end{cases}$$

Write $\operatorname{Fix}\sigma = \{c \in B_n : \sigma(c) = c\}$.

Lemma 6 (`sig_sig`, `sig_eq_iff`, `sig_adj`)**.** *Let $c = (i,j) \in B_n$.*

(1) *$\sigma(\sigma(c)) = c$; in particular σ is injective.*
(2) *$\sigma(c) = c$ if and only if $f(i) \geq n$ and $f(j) \geq n$.*
(3) *If $\sigma(c) \neq c$, then c and $\sigma(c)$ are two steps apart.*

Proof. We go through the three cases of Definition 5. If $f(j) < n$, then $\sigma(c) = (i, f(j))$, a square in the same row whose column differs from j by exactly 2 (Lemma 4); in particular $\sigma(c) \neq c$. Since $f(f(j)) = j < n$, the first case applies to $\sigma(c)$ and gives $\sigma(\sigma(c)) = (i,j)$. If $f(j) \geq n$ and $f(i) < n$, then $\sigma(c) = (f(i), j)$, a square in the same column whose row differs from i by exactly 2, so again $\sigma(c) \neq c$. The column of $\sigma(c)$ is still j, with $f(j) \geq n$, and $f(f(i)) = i < n$, so the second case applies to $\sigma(c)$ and gives $\sigma(\sigma(c)) = (i,j)$. In the third case $\sigma(c) = c$. This proves (1), shows that $\sigma(c) = c$ happens exactly in the third case, which is (2), and proves (3). □

Definition 7 (`P`)**.** The *colour* of $(i,j) \in B_n$ is $\chi(i,j) = (\lfloor i/2 \rfloor + \lfloor j/2 \rfloor) \bmod 2$. Let $S_0 = \{c \in B_n : \chi(c) = 0\}$ and $S_1 = B_n \setminus S_0$.

Thus S_0 is the union of the 2×2 blocks $\{2p, 2p+1\} \times \{2q, 2q+1\}$ (cut off at the border) with $p+q$ even; this is Mathar's checkerboard of 2×2 blocks.

Lemma 8 (`P_flip`, `P_of_fix`)**.** (1) *If $c, d \in B_n$ are two steps apart, then $\chi(c) \neq \chi(d)$.*
(2) *If $\sigma(c) = c$, then $\chi(c) = 0$; that is, $\operatorname{Fix}\sigma \subseteq S_0$.*

Proof. (1) If $c = (i,j)$ and $d = (i, j+2)$, then $\lfloor (j+2)/2 \rfloor = \lfloor j/2 \rfloor + 1$, so the colours differ; the other cases are symmetric. (2) Let $c = (i,j)$. By Lemma 6(2), $f(i) \geq n > i$, so $f(i) = i+2$ and $i \bmod 4 \in \{0,1\}$; hence $\lfloor i/2 \rfloor$ is even. Likewise $\lfloor j/2 \rfloor$ is even, so $\chi(c) = 0$. □

Proposition 9 (`key`)**.** (1) *Every admissible $T \subseteq B_n$ satisfies $2|T| \leq n^2 + |\operatorname{Fix}\sigma|$.*
(2) *S_0 is admissible and $2|S_0| \geq n^2 + |\operatorname{Fix}\sigma|$.*
Consequently $a(n) = |S_0|$ and $2a(n) = n^2 + |\operatorname{Fix}\sigma|$.

Proof. (1) Let T be admissible and $\sigma(T) = \{\sigma(c) : c \in T\}$; then $|\sigma(T)| = |T|$ by Lemma 6(1). We claim $T \cap \sigma(T) \subseteq \operatorname{Fix}\sigma$. Let $c \in T \cap \sigma(T)$, say $c = \sigma(d)$ with $d \in T$. If $\sigma(d) \neq d$, then d and $\sigma(d) = c$ are two squares of T that are two steps apart, by Lemma 6(3), contradicting admissibility. Hence $\sigma(d) = d$, so $c = d$ and $\sigma(c) = c$. By inclusion–exclusion,

$$2|T| = |T| + |\sigma(T)| = |T \cup \sigma(T)| + |T \cap \sigma(T)| \leq n^2 + |\operatorname{Fix}\sigma|.$$

(2) Two squares of S_0 have the same colour, so by Lemma 8(1) they are not two steps apart; thus S_0 is admissible. Let $c \in S_1$. Then $\sigma(c) \neq c$ by Lemma 8(2), so c and $\sigma(c)$ are two steps apart (Lemma 6(3)) and $\chi(\sigma(c)) \neq \chi(c) = 1$, i.e. $\sigma(c) \in S_0$. Moreover $\sigma(c) \notin \operatorname{Fix}\sigma$: otherwise $c = \sigma(\sigma(c)) = \sigma(c)$ would be fixed, and fixed squares lie in S_0. Hence $\sigma(S_1)$ and $\operatorname{Fix}\sigma$ are disjoint subsets of S_0 (Lemma 8(2)), and since σ is injective,

$$n^2 - |S_0| + |\operatorname{Fix}\sigma| = |S_1| + |\operatorname{Fix}\sigma| = |\sigma(S_1)| + |\operatorname{Fix}\sigma| \leq |S_0|.$$

For the consequence: $a(n) \geq |S_0|$ because S_0 is admissible, and every admissible T satisfies $2|T| \leq n^2 + |\operatorname{Fix}\sigma| \leq 2|S_0|$ by (1) and (2), so $a(n) = |S_0|$. Applying (1) to $T = S_0$ and combining with (2) gives $2|S_0| = n^2 + |\operatorname{Fix}\sigma|$. □

Lemma 10 (`card_fix`, `card_A`)**.** *Let* $A_n = \{j \in \{0, \dots, n-1\} : f(j) \geq n\}$. *Then* $\operatorname{Fix}\sigma = A_n \times A_n$, *and*

$$A_n = \emptyset, \quad \{n-1\}, \quad \{n-2, n-1\}, \quad \{n-2\} \qquad \text{for } n \equiv 0, 1, 2, 3 \pmod 4$$

respectively. Hence $|A_n| = 0, 1, 2, 1$ *and* $|\operatorname{Fix}\sigma| = |A_n|^2 = \varepsilon(n)$.

Proof. The first claim is Lemma 6(2). Let $0 \leq j < n$. If $j \bmod 4 \in \{2, 3\}$, then $f(j) = j - 2 < n$, so $j \notin A_n$. If $j \bmod 4 \in \{0, 1\}$, then $f(j) = j + 2$, and $j \in A_n$ if and only if $j \geq n - 2$. So A_n consists of those $j \in \{n-2, n-1\}$ with $0 \leq j$ and $j \bmod 4 \in \{0, 1\}$. The residues of $(n-2, n-1)$ modulo 4 are $(2, 3)$, $(3, 0)$, $(0, 1)$, $(1, 2)$ for $n \equiv 0, 1, 2, 3$, which gives the four cases (for $n = 1$ only $j = 0 = n - 1$ occurs, as claimed). □

Proof of Theorem 2. By Proposition 9 and Lemma 10, $2a(n) = n^2 + |A_n|^2 = n^2 + \varepsilon(n)$ (`two_a`). For $n \equiv 0 \pmod 4$ this gives $a(n) = n^2/2$; for odd n, $a(n) = (n^2 + 1)/2$; for $n \equiv 2 \pmod 4$, $a(n) = (n^2 + 4)/2 = n^2/2 + 2$. In each case the division is exact, so these identities also hold with division in $\mathbb{N}$, as in the Lean statement (`result`). □

Remark 11 (relation to grid graphs)**.** Let G_n be the graph on B_n whose edges join squares two steps apart. By Lemma 8(1) it is bipartite with parts S_0, S_1, and the non-trivial orbits of σ form a matching M with $|M| = (n^2 - \varepsilon(n))/2$. An independent set meets each edge of M at most once, so $a(n) \leq n^2 - |M|$, and Proposition 9 says that S_0 attains this; so M is a maximum matching and S_0 a maximum independent set, as König's theorem predicts for a bipartite graph.

Writing $i = 2p + u$, $j = 2q + v$ with $u, v \in \{0, 1\}$, two squares are two steps apart exactly when they have the same (u, v) and their block coordinates (p, q) are adjacent in the grid. Hence G_n is the disjoint union of four grid graphs $P_r \,\square\, P_s$, $r, s \in \{\lceil n/2 \rceil, \lfloor n/2 \rfloor\}$. In block coordinates, f acts as $q \mapsto q \oplus 1$, so σ tiles each grid by horizontal dominoes, vertical dominoes in the last column when s is odd, and one monomino at a corner when r and s are both odd; and $\chi = (p + q) \bmod 2$ is the checkerboard colouring of each grid. Since $\alpha(P_r \,\square\, P_s) = \lceil rs/2 \rceil$ by exactly this domino-and-checkerboard argument, one can also obtain Theorem 2 by summing $\lceil rs/2 \rceil$ over the four grids: $4\lceil m^2/2 \rceil$ for $n = 2m$, which is $n^2/2$ or $n^2/2 + 2$ according as m is even or odd, and $\lceil (m+1)^2/2 \rceil + 2 \cdot \frac{m(m+1)}{2} + \lceil m^2/2 \rceil = \frac{(2m+1)^2+1}{2}$ for $n = 2m + 1$.

4. Machine verification

Files and declarations. The proof is in the file `lean/Math0/Frontier/OEIS/A348484.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A348484.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A348484.result`, of type `Math0.Statements.A348484.statement`. The auxiliary declarations (all `private`) are those named in Section 3. The definitions of f, σ, χ and the end of the file are, verbatim (docstrings, blank lines, a decidability instance and the lemmas omitted):

```
private def f (j : ℕ) : ℕ := if j % 4 < 2 then j + 2 else j - 2
private def sig {n : ℕ} (c : Fin n × Fin n) : Fin n × Fin n :=
  if h : f c.2 < n then (c.1, ⟨f c.2, h⟩)
  else if h' : f c.1 < n then (⟨f c.1, h'⟩, c.2) else c
private def P {n : ℕ} (c : Fin n × Fin n) : Prop := (c.1.val / 2 + c.2.val / 2) % 2 = 0
theorem result : statement := by
  intro n _
  have h := two_a n
```

```
  have hsq : n ^ 2 = n * n := sq n
  rw [← hsq] at h
  refine ⟨fun h0 => ?_, fun h13 => ?_, fun h2 => ?_⟩
  · rw [if_pos h0] at h; omega
  · rw [if_neg (by omega), if_neg (by omega)] at h; omega
  · rw [if_neg (by omega), if_pos h2] at h; omega
end Math0.Frontier.A348484
#math0_settles Math0.Frontier.A348484.result Math0.Statements.A348484.statement
```

The counting in `key` uses the Mathlib lemmas `Finset.card_image_of_injective`, `Finset.card_union_add_card_inter` (inclusion–exclusion in part (1)), `Finset.card_filter_add_card_filter_not` ($|S_0| + |S_1| = n^2$), `Finset.card_union_of_disjoint` and `Finset.card_le_card`; `card_fix` uses `Finset.card_product`, and `card_A` uses `Finset.card_eq_zero`, `Finset.card_eq_one`, `Finset.card_eq_two`. Arithmetic side goals, including those with mod and division by constants, are closed by the decision procedure `omega`. The proof is uniform in n: it uses no `decide`, no `native_decide`, no `set_option`, and none of the data-check lemmas of the statement file.

Audit. The command `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` use no axioms besides `propext`, `Classical.choice` and `Quot.sound` (this rejects `sorryAx`, `Lean.ofReduceBool` from `native_decide`, and custom axioms); and (3) the type of `result` is syntactically the constant `statement` or its negation, up to metadata, with no unfolding. Here it reports

```
math0 audit: 'Math0.Statements.A348484.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A348484.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A348484.result' proves 'Math0.Statements.A348484.statement'
```

Toolchain and reproduction. We use Lean 4 [3] `v4.33.1` and Mathlib [2] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [1] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. After `lake build Math0.Statements.OEIS.A348484`, the single file can be checked with `lake env lean Math0/Frontier/OEIS/A348484.lean` in the directory `lean` (about 15 seconds on our machine).

References

[1] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An Open and Evolving Benchmark for Verified Discovery in Mathematics*, arXiv:2605.13171, 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[2] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[3] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[4] The OEIS Foundation Inc., *Entry A048716: Numbers k such that (k AND 4 ∗ k) = 0, where AND is bitwise-and* (A. Karttunen, March 30, 1999), The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A048716`.

[5] The OEIS Foundation Inc., *Entry A348484: Maximum number of squares on an n×n chessboard such that no two are two steps apart horizontally or vertically* (Yang Hong, October 20, 2021; revision #33, December 6, 2022), The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A348484`.

Produced by Claude (an AI model developed by Anthropic) in the math0 project